

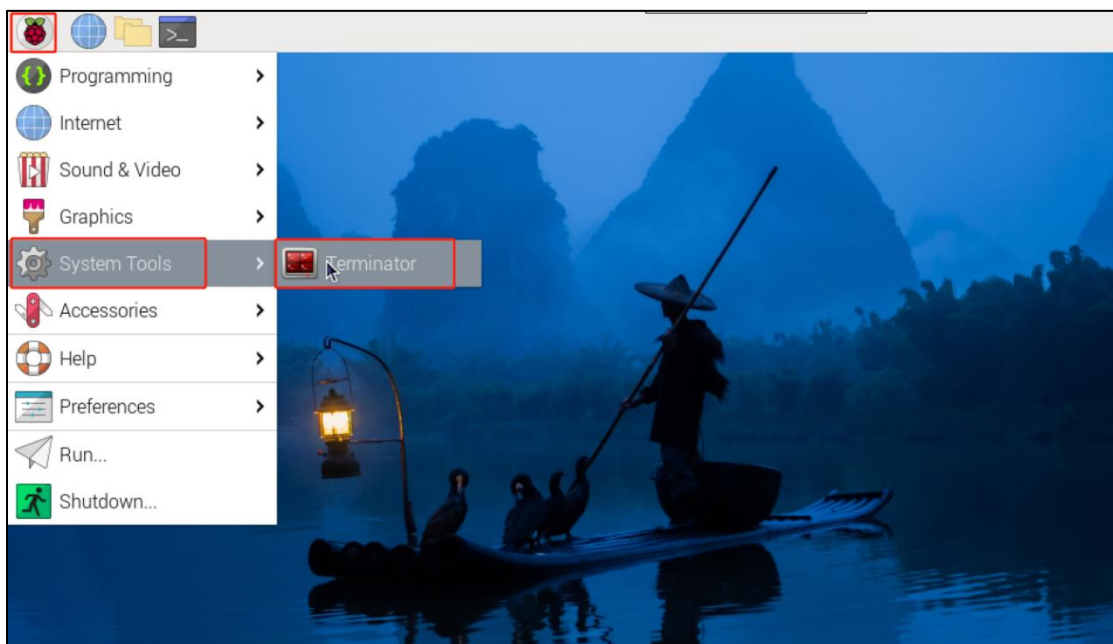
Lesson 5 ROS2 Workspace

1. Introduction to Workspace

In ROS robot development, when we develop specific functionalities for robots, various code, parameters, scripts, and other files need to be placed in a folder for management. This folder is called a workspace in the ROS system. Therefore, a workspace is a folder where project-related files are stored, serving as the main location for storing all data during the development process.

2. Create & Compile Workspace

- 1) Click  in the upper left corner, then select **System Tools** and **Terminator** in sequence.



- 2) Enter “**docker ps**” to list the currently running containers

```

pi@raspberrypi: ~
pi@raspberrypi: ~ 100x24
pi@raspberrypi:~ $ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS        NAMES
55ce8db5f027   ros:humble    "/ros_entrypoint.sh ..." 6 hours ago   Up 4 hours           humble
1318ca2bfbd5   ros:melodic   "/ros_entrypoint.sh ..." 8 hours ago   Up 4 hours           melodic
pi@raspberrypi:~ $

```

3) Enter “**docker exec -it -u ubuntu -w /home/ubuntu 55ce /bin/bash**” (the container ID can be abbreviated as long as it uniquely identifies the container) to enter the contain with ROS2 installed.

```

ubuntu@raspberrypi: ~
ubuntu@raspberrypi: ~ 102x24
pi@raspberrypi:~ $ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS        NAMES
55ce8db5f027   ros:humble    "/ros_entrypoint.sh ..." 7 hours ago   Up 4 hours           humble
1318ca2bfbd5   ros:melodic   "/ros_entrypoint.sh ..." 8 hours ago   Up 4 hours           melodic
pi@raspberrypi:~ $ docker exec -it -u ubuntu -w /home/ubuntu 55ce /bin/bash
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
ubuntu@raspberrypi:~$ █

```

4) Enter “**mkdir -p ~/hiwonder_ws/src**” to create a workspace called “**mkdir -p ~/hiwonder_ws/src**”.

```

ubuntu@raspberrypi: ~
ubuntu@raspberrypi: ~ 102x24
pi@raspberrypi:~ $ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS        NAMES
55ce8db5f027   ros:humble    "/ros_entrypoint.sh ..." 7 hours ago   Up 4 hours           humble
1318ca2bfbd5   ros:melodic   "/ros_entrypoint.sh ..." 8 hours ago   Up 4 hours           melodic
pi@raspberrypi:~ $ docker exec -it -u ubuntu -w /home/ubuntu 55ce /bin/bash
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
ubuntu@raspberrypi:~$ mkdir -p ~/hiwonder_ws/src

```

5) Enter “**cd hiwonder_ws**” to switch to the workspace named “**hiwonder_ws**”.

```

ubuntu@raspberrypi:~$ cd hiwonder_ws/
ubuntu@raspberrypi:~/hiwonder_ws$ █

```

6) Enter “**colcon build**” to compile the workspace.

```
ubuntu@raspberrypi:~/hiwonder_ws$ colcon build  
Summary: 0 packages finished [1.00s]
```

7) Enter “**source ~/hiwonder_ws/install/setup.bash**” to load the workspace environment in ROS2.

```
ubuntu@raspberrypi:~$ source ~/hiwonder_ws/install/setup.bash  
ubuntu@raspberrypi:~$
```

8) Every time you open a new terminal, step 7) will be executed to load the workspace. However, you can enter the command “**echo “source ~/hiwonder_ws/install/setup.bash” >> ~/.bashrc**” to append the specified command to the .bashrc file.

```
ubuntu@raspberrypi:~$ echo "source ~/hiwonder_ws/install/setup.bash" >> ~/.bashrc  
ubuntu@raspberrypi:~$
```

9) Then enter “**source ~/.bashrc**” to make the .bashrc file take effect, thereby there is no need to load the workspace environment for each operation.

```
ubuntu@raspberrypi:~$ source ~/.bashrc  
ubuntu@raspberrypi:~$
```

3. Introduction to Workspace

After compilation, enter the command “**tree -L 1**” to view the root directory of the workspace.

```
ubuntu@raspberrypi:~/hiwonder_ws$ tree -L 1  
.  
├── build  
├── install  
├── log  
└── src
```

A typical workspace structure in ROS system is shown as above. The “hiwonder_ws” is the root directory of the workspace, which contains four sub-directories, or sub-spaces.

Name	Meaning	Instruction
build	Compilation workspace	Save intermediate files generated during the compilation process.
install	Installation workspace	Store the compiled executable files and scripts.
log	Log workspace	Save various logs including warnings, errors, and information during the compilation and execution processes
src	Code workspace	The subsequent code and scripts need to be manually placed here.

All in all,most of operations are performed within the “src” folder among these four spaces. After successful compilation, the results will be executed from the “install” folder. The “build” and “folder” are rarely used.

Moreover, the name of the workspace can be customized, and there can be multiple workspaces. For example:

Workspace 1: hiwonder_ws_a, used for development of robot A.

Workspace2: hiwonder_ws_b, used for development of robot B.

Workspace3: hiwonder_ws_c, used for development of robot C.

The above situations are all permitted, similar to create multiple new projects in an integrated development environment, they all exists in parallel.